

Dr. Chunhua Liu

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As a researcher in natural language processing (NLP), my vision is to develop adaptive and reliable AI systems that enable safe and effective human-AI interactions. I actively engage in interdisciplinary collaboration across cognitive psychology, software engineering, and education.

- **Publications:** Published 9 first-authored papers in tier-1 venues (CORE A/B or 26-29% acceptance rate)
- **Publication Impact:** Received 156 citations and an h-index of 6 (Google Scholar)
- **Research Grant:** Secured internal faculty-supported collaboration funding through the Melbourne Data Analytics Platform (MDAP; acceptance rate $\approx 30\%$)
- **Professional Service:** Served as a referee for 6 CORE rank A*/A/B journals and conferences
- **Research Collaboration:** Co-authored 10 papers with 10 researchers from 6 academic institutions in Singapore, Canada, Australia, China, and UK (on-going)
- **Teaching Experience:** Co-lectured 1 and tutored 4 postgraduate courses
- **Supervision Record:** Co-supervising 1 PhD student; co-supervised 3 Master students

APPOINTMENT & EDUCATION

PostDoc in The University of Melbourne (Level B) Topic: Disinformation/Persuasion in Natural Language Processing Advisor: Professor Eduard Hovy	<i>Jan 2025 - Present</i>
PostDoc in The University of Melbourne (Level A) Topic: Automated Code Review in Software Engineering Advisor: Dr. Patanamon Thongtanunam	<i>Jan 2024 - Dec 2024</i>
Ph.D. in The University of Melbourne Topic: Commonsense Reasoning in Natural Language Processing Supervisors: Prof. Trevor Cohn and Dr. Lea Frermann	<i>Sep 2019 - Nov 2023</i>
M.Sc. in Beijing Language and Culture University, China Major: Computer Application Technology (GPA: 90.2/100) Topic: Natural Language Understanding, Supervisor: A/Prof. Dong Yu	<i>Sep 2016 - Jul 2019</i>
B.Sc. in Beijing Language and Culture University, China Major: Computer Science and Technology (GPA: 85.3/100) Topic: "Character Relationship Mining", Supervisor: A/Prof. Dong Yu	<i>Sep 2012 - Jul 2016</i>

INVITED TALKS AND PRESENTATIONS

Invited Talk at NextGen AI Hackathon, Melbourne, Australia	<i>Aug 2025</i>
Poster Presentation at ACL main, Vienna, Austria	<i>Jul 2025</i>
Demo Multi-Agent Disinfo-Avatar at Information and Influence Lab, 2025, Melbourne, Australia	<i>Jul 2025</i>
Demo Disinformation Avatar at The Future of Disinformation Event, 2025, Melbourne, Australia	<i>Apr 2025</i>
Paper Oral Presentation at MSR 2025, Ottawa, Canada	<i>Apr 2025</i>
Invited Talk in the Vision and Language Group, Monash University	<i>Sep 2023</i>
Paper Poster Presentation in the 44th Cognitive Science Society, Sydney	<i>Jul 2023</i>
Invited Talk at the Computational Linguistics Group, University of Toronto	<i>Jul 2023</i>

Paper Oral Presentation at *SEM 2023, Toronto, Canada	<i>Aug 2023</i>
Invited Talk at Computational Cognitive Science Lab, University of Melbourne	<i>Nov 2022</i>
Paper Poster Presentation at ACL 2022, Online	<i>Aug 2022</i>
Paper Oral Presentation at CoNLL 2021, Online	<i>Aug 2021</i>
Paper Oral Presentation at the NLPCC 2018, Hohhot, China	<i>Aug 2018</i>

TEACHING

Co-lecturer “Software Modelling and Design”	<i>2024</i>
M.Sc course, Sch. of Computing and Information System, UniMelb	
Guest Lecturer “Natural Language Processing”	<i>2024</i>
2nd year M.Sc course, Sch. of Computing and Information System, UniMelb	
Tutor “Text Analytics for Business”	<i>2021,2022</i>
M.B.A course, Melbourne Business School, UniMelb	
Tutor “Introduction to Machine Learning”	<i>2021,2022</i>
1st year M.Sc course, Sch. of Computing and Information System, UniMelb	
Tutor “Natural Language Processing”	<i>2021,2022</i>
2nd year M.Sc course, Sch. of Computing and Information System, UniMelb	
Project Supervisor of Data Science Project	<i>2021</i>
2nd year of M.Ds course, Sch. of Computing and Information System, UniMelb	

RESEARCH SUPERVISION

Software Engineering (with Dr. Patanamon Thongtanunam)	<i>2024-Present</i>
Co-supervising 1 PhD student (Hong Yi Lin) on Automated Code Review (Confirmation completed; TOSEM/ACL submissions)	
Co-supervised 2 Master’s students: Linh Nguyen (Review Comment Classification, Jul-Dec 2024, ICSME submission) and Mukhammad Karimov (Retrieval Augmented Code Review, Feb-Dec 2024)	
Natural Language Processing	<i>Jul-Dec 2024</i>
Co-supervised with Dr. Lea Frermann and Dr. Simon De Deyne one Master’s student (Chaoyi Xiang, Master of IT – AI) on moral values in Western English-speaking societies and LLMs.	
Outcomes: Thesis mark 89/100; Paper at ACL Rolling Review; Student continued to PhD at UniMelb	

PROFESSIONAL SERVICE

Peer Reviewing: ACL ARR (Feb, May 2025; Aug, Feb, Oct 2024), *SEM (2023, 2024, 2025), EMNLP (2023) NLPCC (2018, 2022, 2023), CCL (2018), ACL Student Research Workshop (2023), Australasian Conference on Information Systems (2023), Trustworthy and Reliable Large-Scale Machine Learning Models (2023), Transactions on Software Engineering and Methodology (2024), IEEE Transactions on Software Engineering (2024)	
NLP Reading Group Organizer, Beijing Language and Culture University	<i>2016-2018</i>
Co-organizer of the NLP reading group at The University of Melbourne	<i>2022-2023</i>
Volunteer of the CIPS Summer School, Beijing, China	<i>Jul 2016, 2018</i>

AWARDS AND FUNDING

MDE Discretionary Fund (15K), UniMelb	<i>2025</i>
MDAP faculty supported collaboration, UniMelb	<i>2025</i>
Outstanding Paper Award, NLPCC	<i>2018</i>
National Scholarship (Top 5% at school)	<i>2018</i>
1st Prize Scholarship	<i>2018</i>
National Encouragement Scholarship(Top 5% at school)	<i>2014, 2015</i>

PUBLICATIONS

- [1] **Chunhua, Liu**, Lin Hong Yi, and Thongtanunam Patanamon. “Hallucinations in Code Change to Natural Language Generation: Prevalence and Evaluation of Detection Metrics”. In: *International Joint Conference on Natural Language Processing & Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)*. 2025.
- [2] **Chunhua, Liu**, Lin Hong Yi, and Thongtanunam Patanamon. “Too Noisy To Learn: Enhancing Data Quality for Code Review Comment Generation”. In: *Proceedings of the 22nd International Conference on Mining Software Repositories (MSR)*. 2025.
- [3] Chaoyi Xiang, **Chunhua, Liu**, Simon De Deyne, and Lea Frermann. “Comparing Moral Values in Western English-speaking societies and LLMs with Word Associations”. In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. July 2025.
- [4] Hong Yi Lin and Patanamon Thongtanunam and Christoph Treude and Michael Godfrey and **Chunhua Liu** and Wachiraphan Charoenwet. “Leveraging Reviewer Experience in Code Review Comment Generation”. In: *ACM Transactions on Software Engineering and Methodology (TOSEM)* (2025).
- [5] Linh Nguyen, **Chunhua, Liu**, Lin Hong Yi, and Thongtanunam Patanamon. “Exploring the Potential of Large Language Models in Fine-Grained Review Comment Classification”. In: *Proceedings of the 21st International Conference on Software Composition and Analysis (SCAM)*. 2025.
- [6] Lin Hong Yi, **Chunhua, Liu**, Gao Haoyu, Thongtanunam Patanamon, and Treude Christoph. “CodeReviewQA: The Code Review Comprehension Assessment for Large Language Models”. In: *Findings of the Association for Computational Linguistics: ACL 2025*. July 2025, pp. 9138–9166.
- [7] Simon De Deyne, **Chunhua Liu**, and Lea Frermann. “Can GPT-4 Recover Latent Semantic Relational Information from Word Associations? A Detailed Analysis of Agreement with Human-annotated Semantic Ontologies.” In: *The Workshop on Cognitive Aspects of the Lexicon (CogALex@ LREC-COLING 2024)*. 2024, p. 68.
- [8] **Chunhua Liu**, Trevor Cohn, and Lea Frermann. “Seeking Clozure Robust Hypernym Extraction from BERT with Anchored Prompts”. In: *The 12th Joint Conference on Lexical and Computational Semantics (*SEM)*. 2023.
- [9] **Chunhua Liu**, Trevor Cohn, Simon De Deyne, and Lea Frermann. “WAX: A New Dataset for Word Association eXplanations”. In: *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing*. 2022, pp. 106–120.
- [10] **Chunhua Liu**, Trevor Cohn, and Lea Frermann. “Commonsense Knowledge in Word Associations and ConceptNet”. In: *Proceedings of the 25th Conference on Computational Natural Language Learning*. 2021, pp. 481–495.
- [11] **Chunhua Liu**, Shan Jiang, Hainan Yu, and Dong Yu. “Multi-turn Inference Matching Network for Natural Language Inference **Outstanding Paper Award**”. In: *Natural Language Processing and Chinese Computing (NLPCC)*. 2018.

- [12] **Chunhua Liu**, Yan Zhao, Qingyi Si, and Dong Yu. “Multi-Perspective Fusion Network for Commonsense Reading Comprehension”. In: *Chinese Computational Linguistics and Natural Language Processing Based on Naturally Annotated Big Data*. 2018, pp. 262–274.
- [13] **Chunhua Liu**, Haiou Zhang, Shan Jiang, and Dong Yu. “DEMn: Distilled-Exposition Enhanced Matching Network for Story Comprehension”. In: *Pacific Asia Conference on Language, Information and Computation (PACLIC)*. 2018.
- [14] Meiqian Zhao, **Chunhua Liu**, Lu Liu, Yan Zhao, and Dong Yu. “BLCU_NLP at SemEval-2018 Task 12: An Ensemble Model for Argument Reasoning Based on Hierarchical Attention”. In: *NAACL-HLT 2018 Joint workshop on semantic evaluation (SemEval)*. 2018.
- [15] Yukun Feng, Dong Yu, Jian Xu, and **Chunhua Liu**. “Semantic Frame Labeling with Target-based Neural Model”. In: *6th Joint Conference on Lexical and Computational Semantics (*SEM)*. 2017, pp. 91–96.
- [16] Shan Jiang, Bohan Li, **Chunhua Liu**, and Dong Yu. “Knowledge Augmented Inference Network for Natural Language Inference”. In: *Knowledge Graph and Semantic Computing. Knowledge Computing and Language Understanding*. Singapore, 2019, pp. 129–135.
- [17] Yan Zhao, Lu Liu, **Chunhua Liu**, Ruoyao Yang, and Dong Yu. “From Plots to Endings: A Reinforced Pointer Generator for Story Ending Generation”. In: *Natural Language Processing and Chinese Computing (NLPCC)*. 2018, pp. 51–63.
- [18] Dong Yu, **Chunhua Liu**, and Yue Tian. “Personal title and career attributes extraction based on distant supervision and pattern matching (*In Chinese*)”. In: *Journal of Computing Applications* 36.2 (2016), pp. 455–459.